Shihan Fu

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Ph.D. student in Computer Science at Northeastern University. I build **health AI** systems that combine **wearable passive sensing** with **LLM-based conversational agents**, helping people capture, understand, and reflect on **stress** and well-being in everyday life.

Experience

Sep 2024 – present **Graduate Research Assistant** • UbiWell Lab, Northeastern University

- Built **StressTalk** end to end: an **LLM-based conversational agent** for daily stress check-ins, a **multi-agent sensemaking pipeline** that infers the context of **wearable-detected stress episodes**, a **native iOS app (Swift)**, and a **Python/Flask + MongoDB** backend.
- Ran a **6-week, 21-participant** within-subjects field deployment against **EMA**: **82%** of stressful reports came from sensing-triggered questions, and **15 of 21** participants preferred StressTalk for long-term self-reporting. Analyzed the longitudinal wearable **time-series** (IBI/HRV) and self-report data in **Python** with **linear mixed-effects models**.
- Interviewed **6 domain experts**, showing that conversational self-reports read alongside sensing data clarify stressors and changing appraisals.
- Conducted the **human-centered studies** for clinical AI projects with physicians and ML researchers: early **sepsis prediction** from EHR time series (SepsisCalc, KDD '25) and **cardiotoxicity risk prediction** for cancer patients (CardioAI, CHI '25).
- Building **wearable and voice-assistant** support for caregiver stress, funded by the NIH AI-CARING Institute (demo video).

Jul – Nov 2025 **Applied Scientist Intern** • Amazon, Search Experience Science Group

- Co-authored a **40-participant study** testing whether **LLM-agent digital twins** of real shoppers can evaluate an agentic shopping assistant in their place [P3].
- Built an end-to-end testbed for an **LLM-based conversational shopping assistant** in the **WebArena** simulated e-commerce environment, enabling controlled studies of assistant behavior.
- Translated HCI theories of **consumer decision making** and conversational interaction into concrete **conversation strategies** for the assistant.

Education

Sep 2024 – May 2029 (expected) **Ph.D., Computer Science** • *Northeastern University*
Advisor: Varun Mishra

Sep 2022 – Jul 2024 **M.Phil., Computational Media and Arts** • *The Hong Kong University of Science and Technology*
Advisor: Mingming Fan

Sep 2017 – Jun 2022 **B.A., Russian Language and Literature** • *Zhejiang University*

Skills

Research	HCI, Ubiquitous Computing, Passive Sensing, Health AI, EMA, Mixed Methods
ML / AI	LLM Agents, Multi-Agent Systems, Conversational Agents, Tool Calling, Physiological Signal Processing (HRV), Longitudinal & Mixed-Effects Modeling, PyTorch (coursework)
Engineering	Python, SQL, R, Swift (iOS), Flask, MongoDB
Design	UX Research, Prototyping, Usability Testing

Publications

PREPRINTS & WORK IN PROGRESS

[P1] **StressTalk: Designing and Evaluating a Sensing-Guided Conversational System for Understanding Daily Stress**
Shihan Fu, Ha Le, Akshat Choube, Srinidhi Rajaraman, Jiachen Li, Jixin Li, Vedant Das Swain, and Varun Mishra
Under review, 2026.

[P2] **"It Felt Like I Was Left in the Dark": Exploring Information Needs and Design Opportunities for Family Caregivers of Older Adult Patients in Critical Care Settings**
Shihan Fu, Bingsheng Yao, Smit Desai, Yuqi Hu, Yuling Sun, Samantha Stonbraker, Yanjun Gao, Elizabeth M. Goldberg, and Dakuo Wang
arXiv preprint arXiv:2502.05115, 2025.

[P3] **LLM Agent Meets Agentic AI: Can LLM Agents Simulate Customers to Evaluate Agentic-AI-based Shopping Assistants?**
Lu Sun, **Shihan Fu**, Bingsheng Yao, Yuxuan Lu, Wenbo Li, Hansu Gu, Jiri Gesi, Jing Huang, Chen Luo, and Dakuo Wang
arXiv preprint arXiv:2509.21501, 2025.

[P4] **Human and LLM-based Voice Assistant Interaction: An Analytical Framework for User Verbal and Nonverbal Behaviors**
Szeyi Chan, **Shihan Fu**, Jiachen Li, Bingsheng Yao, Smit Desai, Mirjana Prpa, and Dakuo Wang
arXiv preprint arXiv:2408.16465, 2024.

[P5] **Clinical Challenges and AI Opportunities in Decision-Making for Cancer Treatment-Induced Cardiotoxicity**
Siyi Wu, Weidan Cao, **Shihan Fu**, Bingsheng Yao, Ziqi Yang, Changchang Yin, Varun Mishra, Daniel Addison, Ping Zhang, and Dakuo Wang
arXiv preprint arXiv:2408.03586, 2024.

CONFERENCE PAPERS (PEER-REVIEWED)

[C1] **CardioAI: A Multimodal AI-based System to Support Symptom Monitoring and Risk Prediction of Cancer Treatment-Induced Cardiotoxicity**
Siyi Wu*, Weidan Cao*, **Shihan Fu**, Bingsheng Yao, Ziqi Yang, Changchang Yin, Varun Mishra, Daniel Addison, Ping Zhang, and Dakuo Wang
CHI '25: ACM CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3706598.3714272.

[C2] **SepsisCalc: Integrating Clinical Calculators into Early Sepsis Prediction via Dynamic Temporal Graph Construction**
Changchang Yin, **Shihan Fu**, Bingsheng Yao, Thai-Hoang Pham, Weidan Cao, Dakuo Wang, Jeffrey M. Caterino, and Ping Zhang
KDD '25: ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Applied Data Science Track. DOI: 10.1145/3690624.3709402.

[C3] **Bridging the Literacy Gap for Adults: Streaming and Engaging in Adult Literacy Education through Livestreaming**
Shihan Fu, Jianhao Chen, Emily Kuang, and Mingming Fan
CHI '24: ACM CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3613904.3642423.

[C4] **"It's like having a friend": Exploring the Opportunity of Large Language Models (LLMs) in Assisting Retirees in Their Daily Lives**
Jiaan Li, Yulu Hu, **Shihan Fu**, Zhiqing Wu, Pan Hui, Hailiang Wang, and Mingming Fan
Chinese CHI '24: International Symposium of Chinese CHI. DOI: 10.1145/3758871.3758911.

[C5] **To Reach the Unreachable: Exploring the Potential of VR Hand Redirection for Upper Limb Rehabilitation**
Peixuan Xiong, Yukai Zhang, Nandi Zhang, **Shihan Fu**, Xin Li, Yadan Zheng, Jinni Zhou, Xiquan Hu, and Mingming Fan
CHI '24: ACM CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3613904.3642912.

[C6] **"It is hard to remove from my eye": Design Makeup Residue Visualization System for Chinese Traditional Opera (Xiqu) Performers**
Zeyu Xiong, **Shihan Fu**, Yanying Zhu, Chenqing Zhu, Xiaojuan Ma, and Mingming Fan
CHI '24: ACM CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3613904.3642261 🏆 Honorable Mention (Top 5%).

[C7] **LOOP Meditation: Enhancing Novice's VR Meditation Experience with Physical Movement**
Shihan Fu, Liangliang Qiang, and Wei Zeng
VINCI '23: International Symposium on Visual Information Communication and Interaction. DOI: 10.1145/3615522.3615538.

[C8] **OperARtistry: An AR-based Interactive Application to Assist the Learning of Chinese Traditional Opera (Xiqu) Makeup**
Zeyu Xiong*, **Shihan Fu***, and Mingming Fan
Chinese CHI '23: International Symposium of Chinese CHI. DOI: 10.1145/3629606.3629622 🏆 Honorable Mention (Top 5%).

EXTENDED ABSTRACTS & WORKSHOP PAPERS (PEER-REVIEWED)

[E1] **WEBSERV: A Browser-Server Environment for Efficient Training of Reinforcement Learning-based Web Agents at Scale**
Yuxuan Lu, Jing Huang, Hui Liu, Jiri Gesi, Yan Han, **Shihan Fu**, Tianqi Zheng, and Dakuo Wang
NeurIPS '25 Workshop on Multi-Turn Interactions in Large Language Models.

[E2] **AromaBlendz: An Olfactory System for Crafting Personalized Scents**
Shihan Fu, Jianhao Chen, Yi Cai, and Mingming Fan
CHI EA '24: Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems. DOI: 10.1145/3613905.3648670.

* Equal contribution.

Honors & Awards

2024	*Best Paper Honorable Mention (Top 5%)*, ACM CHI 2024
2023	*Best Paper Honorable Mention (Top 5%)*, Chinese CHI 2023
2022 – 2024	*Postgraduate Studentship (PGS), Full Scholarship*, HKUST

Service

Reviewing	CHI (2025, 2026), CSCW (2025), IMWUT (2024), IJHCS (2024), JMIR (2026)